

The Polynomial Certificate of Transputation: MDL Unification, Quantum Measurement, and the SRRG Fixed Point

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June 2026

<https://doi.org/10.5281/zenodo.20611502>

UGP series hub: <https://doi.org/10.5281/zenodo.20168144>

Classification: Mathematical Physics; Quantum Foundations; Computational Universality; Minimum Description Length.

Abstract

The quantum measurement problem — why a superposition produces a single definite outcome, and why with probability $|c_k|^2$ — remains unresolved across standard interpretations of quantum mechanics, which either postulate collapse, multiply unobservable branches, or explain apparent classicality without resolving outcomes. This paper provides a formal derivation of the measurement mechanism from first principles within the GTE (Generative Triple Evolution) framework.

The GTE polynomial $p(L, C, R) = C + R - CR - LCR$ over $\text{GF}(7)$, the unique Minimum Description Length (MDL)-minimal encoding of the Standard Model generation structure, carries a complete formal certificate of the quantum measurement mechanism. Its Perfect Self-Containment (PSC) projection is f_{MDL} , the discrete adjudicator that defines the measurement orbit $\text{GEN}_1 \rightarrow \text{GEN}_2 \rightarrow \text{GEN}_3 \rightarrow \text{VAC}$ — the only PSC-consistent trajectory in the cellular automaton dynamics — and this projection is characterized in three parts: f_{MDL} agrees with p on the binary sector $\{0, 1\}^3$ (Rule 110), overrides p on the ten SM-orbit neighborhoods to enforce generation-orbit consistency, and projects to zero on all remaining 325 neighborhoods. The polynomial's real diagonal encodes the Self-Referential Renormalization Group (SRRG) adjudication fixed point $\varphi = (\sqrt{5} - 1)/2 \approx 0.618$ (machine-certified, $\mathbf{A}_{\text{Lean}}$). A Three-Level MDL Unification Theorem ($\mathbf{A}/\mathbf{D}$) shows that $P^\top(X) = \arg \min_{x \in X} K(x \mid \text{PSC})$ operates at three nested, non-circular domains: theory selection (p from rule space), field dynamics (Φ_{MDL} via the PMDL variational principle), and quantum event adjudication (Born rule via transputation). The polynomial p is the connecting object: it *is* the output of Level 1, enters the PMDL action at Level 2, and its PSC-projection f_{MDL} defines the orbit used at Level 3. A two-sided computability classification ($\mathbf{A}/\mathbf{D}$; hardness core $\mathbf{A}_{\text{Lean}}$) locates transputation exactly: record-truth on the diagonal fragment is Σ_1^0 -complete, the adjudicator is limit-computable with boundedly many mind changes under a named premise bundle, and hence $\deg_T(P^\top \upharpoonright \text{diag}) = \mathbf{0}'$ exactly — the Gold–Putnam trial-and-error stratum, with no computable convergence modulus. Quantum measurement is trial-and-error computation at the halting degree, and no stronger: physics requires exactly one Turing jump. The same degree- $\mathbf{0}'$ structure reappears at the field level: the irreducible residual of the PMDL action is a left-c.e. real of degree exactly $\mathbf{0}'$, so the cosmological

constant inherits the transputation classification. Differential Self-Adjudicative Computation (DSAC) is positioned as the continuum member of the $[D]$ adjudicator class, bridging the discrete f_{MDL} projection and the transputation selector $P_D^\top$: its relaxation-to-coherence dynamics is the physical realization of the computable stage sequence whose limit is the adjudication outcome.

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Claim-Strength Taxonomy

Every result in this paper carries one of five certification levels:

- **A_{Lean} (A-Lean)** — Lean 4 machine proof, zero **sorry** in **A_{Lean}** results and zero custom axioms in any **A_{Lean}** proof chain. Conditional **A_{Lean}** uses a named physics axiom (always stated explicitly).
- **A/D (A/D)** — Analytic or fully decidable proof; human-verifiable.
- **A (A)** — Computationally verified; reproducible from published scripts.
- **B (B)** — Theoretically motivated conjecture with numerical support.
- **D (D)** — Speculative hypothesis; further development needed.

Ordering: **A_{Lean}** > **A/D** > **A** > **B** > **D**. All Lean certs use **ugp-lean** (canonical public repo) unless stated otherwise; module paths are given in Appendix A.

The Two Levels of the Framework

Level 1 (certificate): $p(L, C, R) = C + R - CR - LCR$ over $\text{GF}(7)$ — the 19-bit polynomial that certifies which structures physics must contain.

Level 2 (field): $\mathcal{L}_{\Phi_{\text{MDL}}} = \frac{1}{2}(\partial_\mu \Phi)^2 - \frac{m^2}{49}(1 - \cos 7\Phi) + \mathcal{L}_{\text{gauge}} + \mathcal{L}_\chi$ — the unique continuum field realising the certificate.

The bridge: $p(x, x, x) = x \Rightarrow m_\varphi = m_\tau$. The Level 2 field law is derived in P42 [22]; full derivation: P48 [13].

1 Introduction

The measurement problem — how quantum superposition becomes a definite outcome — remains unresolved in most interpretations of quantum mechanics. Standard frameworks postulate measurement as an axiom (Copenhagen [1]), multiply branches without a selection mechanism (Everett [4]), rely on decoherence without resolving outcomes [30], or postulate physical collapse [5]. CA-based interpretations of QM also exist [27, 29]. This paper provides a formal derivation of the measurement mechanism from first principles within the GTE framework: quantum measurement is not added as an axiom but is forced — by PSC closure and diagonal self-reference — as a consequence of the universe being a record-bearing, reflexive computational system [13, 23].

The GTE polynomial $p(L, C, R) = C + R - CR - LCR$ over $\text{GF}(7)$ is, from first principles, the unique MDL-minimal CA rule consistent with the Standard Model generation structure and PSC [19, 21]. Three prior papers establish parts of the measurement story: P49 [21] proves the four-object tower $p \rightarrow f_{\text{MDL}} \rightarrow \text{CMCA} \rightarrow \Phi_{\text{MDL}}$ and the invariant-subset uniqueness of f_{MDL} ; P37 [23] derives the full Hilbert space from the f_{MDL} ring and the Born rule from information-theoretic loss; and P48 [13] establishes the GTE transputation framework (D_1 – D_5 constraints, forcing theorems) — the physical realization of the abstract formal theory of transputation [16, 17] — and the MDL three-roles non-circularity argument (P48 §5, “The Three Faces of MDL”).

The present paper establishes what none of those papers state explicitly:

- (i) The **three-part characterization of f_{MDL} as PSC-correction of p** — binary agreement (Rule 110), SM-orbit override, vacuum zeroing — with the information-theoretic gap between p and f_{MDL} computed at both the distributional level (KL divergence: 4.6–7.7 bits) and the algorithmic level (K-gap: 31 bits).

- (ii) The **SRRG fixed-point bridge**: $p_{\mathbb{R}}(x, x, x) = x$ has the positive real root $\varphi = (\sqrt{5} - 1)/2$, identically equal to the SRRG adjudication fixed point — machine-certified (**A_{Lean}**, **gte_poly_srrg_bridge**).
- (iii) The **Three-Level MDL Unification Theorem (A/D)**: $P^{\top}(X) = \arg \min K(x \mid \text{PSC})$ operates with p as the connecting object across all three levels.
- (iv) The **complete discrete measurement mechanism**: from p -dynamics (Class 3 chaotic attractor of period 475) through the PSC-projection π_{PSC} to the f_{MDL} orbit, to transputation as the non-computable continuum realization.
- (v) The **two-sided computability classification of transputation (A/D; hardness core A_{Lean})**: under a named premise bundle (PR1–PR5), record-truth on the diagonal fragment is Σ_1^0 -complete, the adjudicator is limit-computable with mind changes bounded by $2(k_w - 1)$, and $\deg_T(P^{\top} \upharpoonright \text{diag}) = \mathbf{0}'$ exactly — the Gold–Putnam trial-and-error stratum [7, 9] — with an operational no-hypercomputation theorem (no computable convergence modulus). This is the GTE-specific realization of the abstract transputation classification of the NEMS formal theory [17].
- (vi) **DSAC as the [D]-member** bridging the discrete f_{MDL} projection and the non-computable $P_D^{\top}$, now positioned precisely: DSAC’s relaxation dynamics is the physical stage sequence of the classification.

Section 8 locates the GTE transputation mechanism relative to Copenhagen, many-worlds, Bohmian, GRW, and decoherence approaches, explaining why transputation constitutes a structurally distinct fourth option [13]. Section 2.1 (within Section 2) connects the f_{MDL} certificate to Φ_{MDL} via the Algebraic Lifting Theorem, identifying the continuum significance of the three-part characterization.

Reading order. This paper builds on P49 (Objects 0–T, f_{MDL} structure), P37 (Hilbert space and Born rule from the f_{MDL} ring), P48 (transputation framework, MDL three roles, D_1 – D_5 constraints), and P13 [20] (DSAC architecture, TE2.U). Each of those papers is cited rather than reproduced here.

2 The GTE Computational Hierarchy

The GTE framework has a four-level architecture in which each level is the continuum extension or PSC-projection of the level below:

$$\underbrace{p}_{\text{Object 0}} \xrightarrow{\pi_{\text{PSC}}} \underbrace{f_{\text{MDL}}}_{\text{Object 1}} \xrightarrow{\text{CMCA embedding}} \underbrace{\text{CMCA tape}}_{\text{Object 2}} \xrightarrow{\text{Algebraic Lifting}} \underbrace{\Phi_{\text{MDL}}}_{\text{Object 3/T}} . \quad (1)$$

Object 0 is the MDL-minimal polynomial; Object 1 is its PSC-restriction to the admissible orbit; Object 2 embeds Object 1 into a 1+1D chiral CA tape with Rule 110/Rule 124 pair and an asynchronous fractal cellular automaton (AFCA) timing clock; Object T is the continuum field limit. These objects are pairwise distinct: p is a degree-3 polynomial, f_{MDL} is provably non-polynomial (Schwartz–Zippel, **A**), the CMCA tape is a dynamical system in 1+1D, and Φ_{MDL} is a $\mathbb{Z}_7$ -symmetric Klein–Gordon scalar field [21, 25, 26].

The present paper is concerned with the transition $p \xrightarrow{\pi_{\text{PSC}}} f_{\text{MDL}}$ and its extension through the adjudicator class $[D]$ to the quantum measurement selector $P_D^\top$. The transputation layer $[D]$ sits *above* Object T in the adjudication hierarchy:

$$p \rightarrow f_{\text{MDL}} \rightarrow \text{DSAC} \in [D] \rightarrow P_D^\top \text{ (non-computable; degree exactly } \mathbf{0}', \text{ Section 5.3.1).} \quad (2)$$

The CMCA and Φ_{MDL} levels are the *physical substrate*; the $[D]$ -layer is the *adjudication mechanism* operating on top of that substrate. For the full architecture, see P48 [13].

2.1 Connection to Φ_{MDL} via Lifting and Descent

The Algebraic Lifting Theorem ($\mathbf{A}_{\text{Lean}}$, P43 [14]) maps the CMCA algebraic structure (Level 2) to Φ_{MDL} (Level 3) exactly; the Algebraic Descent Theorem ($\mathbf{A}_{\text{Lean}}$, P48 [13]) confirms that the same algebraic structure persists at every finite resolution. The discrete measurement certificate established in this paper therefore has a direct continuum interpretation. The three-part f_{MDL} characterization — binary agreement, orbit override, vacuum zeroing — lifts to a field-theoretic statement: the PSC-admissible kink sector of Φ_{MDL} carries the same three-part structure at continuum scale. The discrete transputation event $\text{GEN}_1 \rightarrow \text{GEN}_2 \rightarrow \text{GEN}_3 \rightarrow \text{VAC}$ lifts to a Φ_{MDL} field-configuration transition: a BPS kink evolving through its three-generation cascade before settling to vacuum. The non-computability of Level 2 $\rightarrow$ 3 lifts to the statement that the Φ_{MDL} path integral’s branch selection is non-computable — the formal content of the D_3 constraint — with its exact strength located by the two-sided classification of Section 5.3.1.

3 f_{MDL} as PSC-Correction of p

3.1 Three-Part Characterization

Objects 0 and 1 are not the same function. Their relationship is a three-part structure:

Three-Part Characterization of f_{MDL}

Part A — Binary sector $\{0, 1\}^3$, 8 inputs ($\mathbf{A}_{\text{Lean}}$):

$f_{\text{MDL}}(L, C, R) = p(L, C, R)$ for all $(L, C, R) \in \{0, 1\}^3$. Both give the Rule 110 truth table [2]. Machine-certified: `p_poly_agrees_fmdl_on_binary` (`UgpLean/Universality/Z7InvariantSubsets.lean`).

Part B — SM orbit neighborhoods, 10 inputs ($\mathbf{A}_{\text{Lean}}$):

f_{MDL} overrides p with PSC-forced orbit values. Witness: $p(1, 1, 5) = 3$ but $f_{\text{MDL}}(1, 1, 5) = 2$ (enforces $\text{GEN}_1 \rightarrow \text{GEN}_2$ winding transition). Machine-certified: `p_fmdl_disagree_on_orbit` (`UgpLean/Polynomial/PolyExplorations.lean`).

Part C — Remaining 325 neighborhoods ($\mathbf{A}_{\text{Lean}}$):

$f_{\text{MDL}}(L, C, R) = 0$ (vacuum projection). Machine-certified: `fmdl_zero_on_free_neighborhoods` (`UgpLean/Universality/CUP3DUniqueness.lean`).

Physical meaning of Part B (orbit override). The ten SM orbit neighborhoods where $f_{\text{MDL}} \neq p$ are precisely the inputs that carry the Standard Model’s generational structure. The raw polynomial does not encode this structure: at $(1, 1, 5)$, for example, p outputs 3 where the SM generation cascade requires 2. The f_{MDL} correction at these inputs is the PSC content: the CA learns to output what the Standard Model’s generation cascade requires, not what the raw

polynomial gives. This correction is not arbitrary — it is uniquely forced by the PSC orbit constraint; no other correction produces the $\text{GEN}_1 \rightarrow \text{GEN}_2 \rightarrow \text{GEN}_3 \rightarrow \text{VAC}$ orbit.

The three parts combine into a single certified statement:

Theorem 3.1 ($\mathbf{A}_{\text{Lean}} - \text{psc_projection_gives_fmdl}$). *f_{MDL} is the PSC-projection of p : it equals p on the binary ether sector, equals the PSC-forced orbit values on the ten SM-orbit neighborhoods, and equals zero on all other inputs. Equivalently, for any $(L, C, R) \notin \{0, 1\}^3 \cup \{\text{SM orbit nbhds}\}$: $f_{\text{MDL}}(L, C, R) = 0$.*

Proof (machine-assisted). The bundle `psc_projection_gives_fmdl` assembles `fmdl_zero_on_free_neighborhoods` (Part C; all $7^3 = 343$ inputs verified by `decide`) and `p_poly_agrees_fmdl_on_binary` (Part A). Part B follows from `p_fmdl_disagree_on_orbit` as a witness. Module paths are listed in Appendix A. $\square$

Remark 3.2. f_{MDL} is *not* p restricted to a subset — it is p *corrected* on the SM orbit to enforce PSC consistency, then zero elsewhere. The correction encodes the PSC content: f_{MDL} is what p “should have been” to satisfy the physical generation-orbit constraint.

3.2 Information-Theoretic Gap

The PSC-projection $p \rightarrow f_{\text{MDL}}$ has two distinct information-theoretic signatures.

Algorithmic complexity gap ($\mathbf{A}_{\text{Lean}}, \mathbf{A}$). The MDL description of p requires 19 bits (`K_CA_CMCA`, $\mathbf{A}_{\text{Lean}}$; the family-level MDL optimality of the GF(7) substrate is `mdl_ca_rule_coding_closed`, $\mathbf{A}_{\text{Lean}}$, P49) [21]. The MDL description of f_{MDL} requires approximately 50 bits (the 8-entry Rule 110 table plus the 10 orbit-override entries, each with position and value). The algorithmic K-gap is $K(f_{\text{MDL}}) - K(p) \approx 31$ bits [8, 10]: the PSC orbit specification is more expensive to encode than the polynomial itself.

Distributional KL divergence ($\mathbf{A}$). Treating p and f_{MDL} as probability distributions over $\text{GF}(7)^3$ (normalized by the sum of positive outputs), the KL divergence is:

$$\text{KL}(\tilde{f}_{\text{MDL}} \parallel \tilde{p}) \approx 4.6\text{--}7.7 \text{ bits}, \quad (3)$$

depending on the model for the SM-orbit entries (Model A: binary only; Model B: binary plus orbit windows). The reverse divergence $\text{KL}(\tilde{p} \parallel \tilde{f}_{\text{MDL}}) \approx 24\text{--}25$ bits reflects the much higher entropy of p as a distribution.

The KL divergence and K-gap are not proportional: KL measures distributional discrepancy at the Shannon level [3, 11] (how different the output histograms are over $\text{GF}(7)^3$), while the K-gap measures the structural overhead of specifying *which* inputs are PSC-admissible. Both quantify aspects of the $p \rightarrow f_{\text{MDL}}$ gap, but at fundamentally different levels of description.

The distributional non-equivalence is machine-certified:

Theorem 3.3 ($\mathbf{A}_{\text{Lean}} - \text{kl_divergence_fmdl_p_nonzero}$). *There exists a triple $(L, C, R) \in \text{GF}(7)^3$ such that $p(L, C, R) \neq f_{\text{MDL}}(L, C, R)$, certifying that p and f_{MDL} are distinct as distributions over $\text{GF}(7)^3$.*

3.3 The Attractor Structure of p

Under iteration of p on the $\mathbb{Z}_7^5$ ring (the 5-cell periodic ring encoding the SM generation winding triples), the dynamics have a precise bifurcation:

Theorem 3.4 (`ALean — period_475_returns, period_475_is_minimal`). *The map p acting on $\text{GF}(7)^5$ has an attractor of minimal period exactly 475. The vacuum basin contains 52 states (0.31% of the 16,807-state space); 99.69% of states asymptote to the period-475 attractor.*

In contrast, under f_{MDL} , the SM generation orbit $\text{GEN}_1 \rightarrow \text{GEN}_2 \rightarrow \text{GEN}_3 \rightarrow \text{VAC}$ is the *unique* attractor:

Theorem 3.5 (`ALean — fmdl_z7_three_generation_orbit`). *The restriction of f_{MDL} to the 5-cell ring produces the three-generation orbit $\text{GEN}_1 \rightarrow \text{GEN}_2 \rightarrow \text{GEN}_3 \rightarrow \text{VAC}$, encoding the $\mathbb{Z}_7$ winding sequence of the Standard Model fermion generations.*

Theorem 3.6 (`ALean — fmdl_gen1_is_garden_of_edden`). *$\text{GEN}_1 = [1, 5, 2, 2, 1]$ is a Garden of Eden under f_{MDL} : it has no predecessor under the f_{MDL} rule. This makes GEN_1 the unique “first cause” of the physical orbit and explains why it is the lightest (most arithmetically stable) SM generation.*

The contrast between p and f_{MDL} is therefore stark: p exhibits Class 3 chaotic dynamics with a period-475 attractor; f_{MDL} has a unique, directed, 3-step orbit with no predecessors at its start. The PSC-projection selects the only trajectory that is physically realizable.

4 The SRRG Fixed-Point Bridge

The GTE polynomial over $\mathbb{R}$ (extending p from $\text{GF}(7)$ to $\mathbb{R}$ by the natural embedding) yields the real-valued function $p_{\mathbb{R}}(L, C, R) = C + R - CR - LCR$. Its diagonal restriction is:

$$p_{\mathbb{R}}(x, x, x) = 2x - x^2 - x^3. \quad (4)$$

The fixed-point equation $p_{\mathbb{R}}(x, x, x) = x$ reduces to:

$$x^3 + x^2 - x = 0 \implies x(x^2 + x - 1) = 0, \quad (5)$$

with roots $x = 0$, $x = (\sqrt{5} - 1)/2 \approx 0.618$, and $x = -(1 + \sqrt{5})/2 \approx -1.618$. The unique positive real root is

$$\varphi = \frac{\sqrt{5} - 1}{2} \approx 0.618034. \quad (6)$$

This value is identically the *SRRG adjudication fixed point*: the Self-Referential Renormalization Group (P27 [24]) identifies the self-consistent running-coupling fixed point $g^* = (\sqrt{5} - 1)/2$ (equivalently $1/\varphi_{\text{large}}$ with $\varphi_{\text{large}} = (1 + \sqrt{5})/2 \approx 1.618$).

Theorem 4.1 (`ALean — gte_poly_srrg_bridge`). *The unique positive real fixed point of $p_{\mathbb{R}}(x, x, x) = x$ is $\varphi = (\sqrt{5} - 1)/2$, identically equal to the SRRG adjudication fixed point g^* . Machine-certified in *srrg-lean* (*SRRGCABridge.lean*, zero sorry).*

Remark 4.2 (Physical significance of Theorem 4.1). The polynomial p — derived from MDL minimality and PSC without reference to continuous dynamics — already encodes the adjudication attractor of the renormalization flow. The real diagonal of p does not merely “accidentally” produce φ : φ is the unique value at which MDL-minimal theory selection (Level 1 of the hierarchy) and the continuum running-coupling fixed point (Level 2) agree. This is the first of two places where the polynomial bridges levels; the second is the Born-rule connection established in Section 5.

Remark 4.3 (The master quadratic). The quadratic $x^2 + x - 1$ in Theorem 4.1 is the master quadratic of the golden ring $\mathbb{Z}[\varphi]$ (P49 [21], golden-quadratic duality): one $\mathbb{Z}$ -quadratic with two completions. Its archimedean fiber — the positive real root φ — is the algebraic input to the discrete-to-continuum SRRG bridge above; its $\text{GF}(7)$ -inertness (the discriminant 5 is a quadratic non-residue mod 7, so the quadratic has no root in $\text{GF}(7)$ or in any 7-adic completion) is precisely the QNR floor that forces the unique binary computational floor of the GTE alphabet. The two fibers are claims at different levels: the real root feeds the continuum bridge, while the $\text{GF}(7)$ -rootlessness is an arithmetic fact about the polynomial itself (Object 0), and the two must not be conflated.

5 Three-Level MDL Unification Theorem

5.1 Statement

The MDL three-roles framework is established in P48 [13] and machine-certified as `mdl_tower_bundle` and `mdl_tower_three_levels_non_circular` (`MDLTower.lean`, `ugp-lean`, $\mathbf{A}_{\text{Lean}}$). The present theorem is distinct: it identifies the *polynomial p itself as the connecting object* that unifies all three levels through a single arithmetic formula.

Theorem 5.1 (Three-Level MDL Unification, $\mathbf{A}/\mathbf{D}$). *The formula*

$$P^\top(X) = \arg \min_{x \in X} K(x \mid \text{PSC constraints}) \quad (7)$$

operates at three nested, logically non-circular levels, with p as the connecting object:

<i>Level</i>	<i>Transition</i>	<i>Domain X</i>	<i>Result</i>	<i>Cert</i>
$0 \rightarrow 1$	<i>Theory selection</i>	<i>All $\mathbb{Z}_N$ CA rules</i>	<i>p (19-bit polynomial, MDL-unique)</i>	$\mathbf{A}_{\text{Lean}}$
$1 \rightarrow 2$	<i>Field dynamics</i>	<i>Φ_{MDL} field configurations</i>	<i>PMDL variational solution; $\nabla^2 \Phi_{\text{MDL}} =$ $G_{\text{eff}} p(w_x, w_y, w_z)$</i>	$\mathbf{A}/\mathbf{D}$
$2 \rightarrow 3$	<i>Event adjudication</i>	<i>Realizations consistent with record w</i>	<i>Born rule $P(k) = c_k ^2$ via $P_D^\top$</i>	$\mathbf{A}_{\text{Lean}}$

The polynomial p connects all three levels: p is the output of Level $0 \rightarrow 1$; p enters the PMDL action at Level $1 \rightarrow 2$ as the coupling functional; and p 's PSC-projection f_{MDL} defines the orbit used by $P_D^\top$ at Level $2 \rightarrow 3$.

The PMDL action entering at Level $1 \rightarrow 2$ additionally carries an irreducible adjudication residual D_{res} ; its computability classification is given in Remark 5.14.

Proof sketch (CatAD). Level $0 \rightarrow 1$ ($\mathbf{A}_{\text{Lean}}$). The MDL selection chain is certified at two quantifier levels. At the family level, $\mathbb{Z}_7 \times \mathbb{Z}_3$ is the MDL-minimal $\mathbb{Z}_N \times \mathbb{Z}_M$ substrate: its combined structure-plus-orbit-data coding cost is strictly below that of the $\mathbb{Z}_5 \times \mathbb{Z}_3$ alternative, whose lack of PSC-admissible kink orbits is verified exhaustively (`mdl_ca_rule_coding_closed`, $\mathbf{A}_{\text{Lean}}$). Within the $\text{GF}(7)$ class, exactly one of the 7^8 multilinear rules satisfies the generation-orbit ring evaluations together with vacuum transparency, and it is p (`ugp_orbit_interpolation_lift`, $\mathbf{A}_{\text{Lean}}$). The

description length $K_{\text{CA}} = 19$ bits of the selected construction is itself machine-certified ($\mathbf{K_CA_CMCA}$, $\mathbf{A_{Lean}}$; see P49 Theorem T96-02 [21]).

Level 1 $\rightarrow$ 2 ($\mathbf{A/D}$). The PMDL action $S_{\text{PMDL}} = \int \Phi_{\text{MDL}} \cdot p(w_x, w_y, w_z) d^3x$ is the unique MDL-minimal action with $\mathbb{Z}_7$ symmetry and the CMCA algebraic certificate; its variational equation yields $\nabla^2 \Phi_{\text{MDL}} = G_{\text{eff}} \cdot p(w_x, w_y, w_z)$ ($\mathbf{A/D}$ in the continuum, via the Algebraic Lifting Theorem; the uniqueness of the cubic coupling is machine-certified, `unique_cubic_gravity_coupling`, $\mathbf{A_{Lean}}$; see P44–P47).

Level 2 $\rightarrow$ 3 ($\mathbf{A_{Lean}}$). PSC closure + diagonal self-reference force $P_D^\top$ as the internal adjudicator (`closed_choice_forces_transputation`, `transputation-lean`, zero custom axioms in this proof chain, [13]). The D_5 constraint forces Born-rule consistency: $P(k) = |c_k|^2$ (`born_rule_unconditional`, `UgpLean/Universality/PSCEffectMeasure.lean`, $\mathbf{A_{Lean}}$). The realizations are defined over the f_{MDL} orbit, connecting this level to the Level 0 $\rightarrow$ 1 output.

Non-circularity. The three domains are strictly nested: theory space $\supset$ field configurations $\supset$ candidate realizations. No application of (7) at one level refers to any object at another level. The non-circularity is machine-certified: `mdl_tower_three_levels_non_circular` (`MDLTower.lean`, `ugp-lean`, $\mathbf{A_{Lean}}$; see P48 [13]). $\square$

5.2 Bundle Certificate

Bundle Certificate ($\mathbf{A_{Lean}}$, zero sorry): The cross-module bundle `mdl_three_level_unification` (`Polynomial/MDLThreeLevelUnification.lean`, `ugp-lean`) certifies the Three-Level Unification at the machine-verified level, wrapping:

- Level 0 $\rightarrow$ 1: `mdl_ca_rule_coding_closed` ($\mathbf{A_{Lean}}$, `MDLDerivabilityCriterion.lean`);
- Level 1 $\rightarrow$ 2: `gte_poly_srrg_bridge` ($\mathbf{A_{Lean}}$, `SRRGCABridge.lean`);
- PSC link $p \rightarrow f_{\text{MDL}}$: `psc_projection_gives_fmld` ($\mathbf{A_{Lean}}$, `PolyExplorations.lean`);
- Level 2 $\rightarrow$ 3 measurement-orbit uniqueness — `fmld_orbit_is_unique_psc_trajectory` ($\mathbf{A_{Lean}}$; `CUP3DUniqueness.lean`).

Level 2 $\rightarrow$ 3 adjudication forcing under PSC and record-divergent choice is certified separately by `closed_choice_forces_transputation` ($\mathbf{A_{Lean}}$; `transputation-lean`, via `mdl_level23_closed_choice_forces_transputation` in the same module).

Corollary 5.2 (Selector Member-Independence, $\mathbf{A/D}$). *The Three-Level MDL Unification Theorem implies that the adjudication selector $P_D^\top$ at Level 2 $\rightarrow$ 3 is uniquely determined by the MDL functional governing Levels 0 $\rightarrow$ 1 and 1 $\rightarrow$ 2. Combined with the D_4 – D_5 uniqueness constraints (§6.1), any two members $D, D' \in [D]$ satisfying D_1 – D_5 induce the same selector on all PSC-admissible adjudication records. Formally: the D_1 – D_5 axioms force the selector to the Born-rule state (`d1_d5_forces_gibbs_family`, `c2_selector_member_independent`, both $\mathbf{A_{Lean}}$), establishing selector member-independence — the C2 property — as a derived theorem. This is the strongest corollary of the unification theorem: not only is there a single MDL functional at all three levels, but physical outcome selection is insensitive to which representative of the class $[D]$ is realized.*

5.3 The Computability Gap

The three levels differ in their computational character:

Proposition 5.3 ($\mathbf{A/D}$). *Level 0 $\rightarrow$ 1 (selecting p from rule space) is computable: the PSC-restriction π_{PSC} on the finite space of $\mathbb{Z}_7$ -state CA rules is an effective decision procedure. Level 2 $\rightarrow$ 3 (transputation $P_D^\top$) is non-computable: the D_3 constraint forces a diagonal barrier that no total-effective algorithm can resolve on the self-referential fragment of the record.*

Proof. Computability of Level $0 \rightarrow 1$: the space of 7-state 3-input CA rules has $7^7 = 7^{343}$ elements; PSC filtering is an explicit finite enumeration; the MDL minimum is attained at p . Non-computability of Level $2 \rightarrow 3$: `asr_rt_not_computable` certifies that no total computable function can decide record-truth on the Arithmetic Self-Reference fragment (`nems-lean`, `Diagonal/HaltingReduction`, `A_Lean`; P48 [13]). $\square$

5.3.1 The Two-Sided Classification

Proposition 5.3 is a lower bound only: it says transputation exceeds total-effective computation, but does not say by how much, leaving open the misreading that transputation is hypercomputation. The following classification closes the gap from both sides. It locates the adjudication function exactly: at the halting degree $\mathbf{0}'$, in the Gold–Putnam “trial-and-error” stratum [7, 9], and no higher. The abstract form of this result — that a closed, record-bearing, diagonal-capable system forces an internal adjudicator whose decision content sits at degree $\mathbf{0}'$ and is realized as a Putnam–Gold trial-and-error limit — is the subject of the NEMS formal theory of transputation [17]; the theorems below are its GTE-specific physical realization, with DSAC as the concrete $[D]$ -member ([16], §6).

The classification concerns the adjudication layer $[D]$, not the substrate levels below it: the record fragment and the finite winding-sector adjudication are defined over f_{MDL} (Object 1, the discrete adjudicator), and $P_D^\top/\text{DSAC}$ is the continuum $[D]$ -member operating above Φ_{MDL} (Object 3/T). The degree statements below are properties of the adjudication function over these levels — not of the polynomial p (Object 0) or of the CMCA tape architecture (Object 2).

The premise bundle. The upper bound is conditional on five named structural premises. They are explicit hypotheses about the GTE record theory — not hidden assumptions — and each names a specific attack surface for refutation.

Definition 5.4 (Premise bundle PR1–PR5). **PR1 (Finite-witness record truth):** *record-truth on the diagonal fragment is Σ_1^0 : every record-inscribed fact has a finite witness. Physics writes only Σ_1 sentences into records — a finite stable configuration certifies an occurrence, but no finite record certifies a non-occurrence over unbounded time. (The framework’s own physical instantiation: vacuum reachability under f_{MDL} is Σ_1^0 -complete, P34 [18].)*

PR2 (Effective candidate enumeration): *a Turing machine enumerates the admissible continuations $S(w)$ of a record w . This is constitutive of the transputation class as defined in P34 [18].*

PR3 (Finite sector structure with MDL-conservativity): *(a) the record-equivalence quotient of $S(w)$ is finite — adjudication outcomes are winding-sector labels drawn from the finite PSC-admissible list; (b) dissonance comparisons are computable on enumerated candidates; (c) among unrefuted candidates, positing an unwitnessed event strictly raises the description cost — the operational content of $P^\top(X) = \arg \min K(x \mid \text{PSC})$.*

PR4 (Π_1 admissibility / Σ_1 refutation): *candidate–record inconsistency manifests at a finite stage; admissibility is Π_1^0 .*

PR5 (Record readout): *at diagonal choice points, record-distinct branches are computably labeled by the coded statement’s truth bit.*

The refutation surface is exactly: physically inscribed Π_2 facts (against PR1) or infinite adjudication sectors (against PR3(a)).

Hardness. The lower bound needs no new mathematics: the machine-certified halting bridge of the Arithmetic Self-Reference structure is already a many-one reduction.

Theorem 5.5 (Σ_1^0 -completeness of diagonal record-truth; $\mathbf{A}/\mathbf{D}$, hardness core $\mathbf{A}_{\text{Lean}}$). *Under the Arithmetic Self-Reference structure and PR1, record-truth restricted to the diagonal fragment, $\text{RT} \upharpoonright \text{diag}$, is Σ_1^0 -complete under many-one reduction: it is many-one equivalent to the halting problem, lies in $\Delta_2^0 \setminus \Delta_1^0$, has Turing degree exactly $\mathbf{0}'$, and is 1-c.e. (one mind change suffices: guess “false,” switch when the witness arrives, never switch back). The hardness direction is machine-certified unconditionally (`halting_manyOne_reducible_to_RT`, $\mathbf{A}_{\text{Lean}}$); the membership direction is machine-certified with PR1 as a named hypothesis structure (`rt_manyOne_reducible_to_halting_zero`, `rt_sigma1_complete_on_diagonal`, $\mathbf{A}_{\text{Lean}}$; `nems-lean`).*

Upper bound.

Theorem 5.6 (Limit computability of the adjudicator; $\mathbf{A}/\mathbf{D}$ conditional on PR1–PR4). *Under PR1–PR4, with $D \in [D]$ satisfying D_1 – D_5 , the sector-valued adjudicator is limit-computable: there is a total computable stage function g with*

$$P_D^\top(w) = \lim_{s \rightarrow \infty} g(w, s), \quad (8)$$

and the number of mind changes (stages s with $g(w, s+1) \neq g(w, s)$) is at most $2(k_w - 1)$, where k_w is the number of admissible winding sectors of the record w . For single-kink adjudication events ($k_w \leq 5$), at most 8 mind changes occur. By the Shoenfield limit lemma [12], $P_D^\top \leq_T \emptyset'$: transputation never exceeds the first Turing jump. The stage construction is machine-certified with the stage-convergence hypothesis as a named structure (`pt_limit_computable`, `transputation-lean`; see Appendix A for the disclosure of its conditionality).

Proof sketch. By PR2 and PR4 there are computable predicates $\text{alive}(w, i, s)$ (non-increasing in s) and $\text{wit}(w, i, s)$ (Σ_1 witness flags), obtained by running the computable dynamics s steps against each candidate’s asserted record. Define $g(w, s)$ as the dissonance-argmin over the alive candidates enumerated by stage s , with MDL-conservativity (PR3(c)) charging unwitnessed posited events. Since k_w is finite (PR3(a)), there is a stage after which all sectors have enumerated and all arriving witnesses have arrived; sector-distinguishing facts are Σ_1 (PR1), so candidates omitting an arrived witness are refuted, while candidates positing a never-arriving witness are never D -minimal. The conservative argmin is therefore eventually constant and equals the unique D_4 -minimum. Each mind change is a new-sector arrival or a refutation/witness flip, giving the $2(k_w - 1)$ bound. The MDL-conservativity premise carries the bound in the asymmetric case: refutation of the positing candidate is Π_1 and never fires at a finite stage; it is the $\arg \min K(x \mid \text{PSC})$ structure itself that keeps the approximation on the conservative branch and makes the limit correct. $\square$

Completeness.

Theorem 5.7 (Degree completeness; $\mathbf{A}/\mathbf{D}$ conditional on PR5). *Under the Arithmetic Self-Reference structure and PR5, $\emptyset' \leq_T P_D^\top \upharpoonright \text{diag}$. Combined with Theorems 5.5 and 5.6:*

$$\deg_T(P_D^\top \upharpoonright \text{diag}) = \deg_T(\text{RT} \upharpoonright \text{diag}) = \mathbf{0}' \quad \text{exactly.} \quad (9)$$

The adjudication function on the diagonal fragment is Δ_2^0 -Turing-complete, in the bounded-mind-change stratum: quantum measurement is trial-and-error computation in Putnam’s exact sense [9], equivalently Gold limiting recursion [7] — at the halting degree, and no stronger. The many-one

form is machine-certified (*pt_diagonal_manyOne_degree_halting*, $\mathbf{A}_{\text{Lean}}$, *transputation-lean*); the bundled Turing-degree form carries its stage-convergence and mind-change hypotheses as named structures (*pt_diagonal_degree_halting*; Appendix A).

Proof sketch. PR5 labels diagonal branches by the coded statement’s truth bit; D_2 (only record-true completions are selectable) forces the label of $P_D^\top(w_n)$ to equal $\chi_{\text{RT}}(n)$. Hence $\text{RT} \upharpoonright \text{diag} \leq_T P_D^\top \upharpoonright \text{diag}$; chain with $\emptyset' \leq_m \text{RT} \upharpoonright \text{diag}$ (Theorem 5.5). $\square$

Operational consequence.

Theorem 5.8 (No certified hypercomputation from measurement; $\mathbf{A}_{\text{Lean}}$ anchor, unconditional). *No computable modulus of convergence exists for the stage sequence g : a computable stage bound $s_0(n)$ would make $\chi_{\text{RT}}(n) = g(n, s_0(n))$ computable, contradicting *asr_rt_not_computable* ($\mathbf{A}_{\text{Lean}}$). Consequently “adjudication complete” is not an internal observable, and no certified Σ_1^0 decision is extractable from finitely many physical events. Machine-certified: *no_computable_convergence_modulus* ($\mathbf{A}_{\text{Lean}}$, *nems-lean*).*

Putnam’s observation that a trial-and-error solver is “never sure” it has the correct answer [9] is here upgraded to a theorem about physical measurement: nature converges to the right adjudication outcome after finitely many revisions, but no physical protocol can certify, at any finite stage, that convergence has occurred.

Corollary 5.9 (One-jump universe; $\mathbf{A}/\mathbf{D}$ conditional on PR1–PR5). *Physics requires exactly one Turing jump. The adjudication function is strictly above degree $\mathbf{0}$ (Proposition 5.3) and exactly at degree $\mathbf{0}'$ (Theorem 5.7); physics never requires $\emptyset''$. The cap is enforced by the record theory itself: realized adjudication outcomes are ordinary finite records (Σ_1 facts about realized history, so PR1 applies to them in turn), while Π_2 counterfactual-limit statements are never physically inscribed — jump iteration through nested adjudication is blocked. This is falsifiable in principle: any physical process of Turing degree strictly above $\mathbf{0}'$ refutes the GTE adjudication architecture.*

Remark 5.10 (Bounded mind changes). The classification places transputation in the bounded-mind-change (Ershov) stratum: diagonal record-truth is 1-c.e., diagonal selection needs at most one revision, and compound single-kink adjudication needs at most $2(k_w - 1) \leq 8$. Transputation is simultaneously as informative as the halting problem (degree $\mathbf{0}'$) and as tame as a finitely-revisable guesser — the precise mathematical form of “lawful but non-algorithmic.” (There is no tension: $\emptyset'$ itself is 1-c.e.) The stage-convergence structure is verified computationally ($\mathbf{A}$): the diagonal family converges with exactly one mind change at the halting time; 500/500 compound-record trials converge within the bound (maximum observed: 3 mind changes); and convergence stages exhibit the heavy-tailed, modulus-free phenomenology of Theorem 5.8 (Appendix B).

Remark 5.11 (Thermodynamic cost of transputation). The mind-change bound $2(k_w - 1) \leq 8$ converts directly into a machine-certified thermodynamic prediction via Landauer’s principle (*landauer_heat_floor_transputation*, *landauer_heat_ceiling_transputation*, $\mathbf{A}_{\text{Lean}}$; *Measurement/LandauerFloor*, *ugp-lean*). With $k_w = 5$ PSC-admissible kink sectors, each single-kink transputation event erases $\ln 5$ bits of alternative winding at cost $Q_{\min} = k_B T \ln 5 \approx 1.61 k_B T$, giving a floor on the thermodynamic heat released per adjudication event. The full $2(k_w - 1) = 8$ mind-change budget gives the ceiling $Q_{\max} = 8 k_B T \ln 2$. Both bounds are falsifiable: any single-kink adjudication event dissipating less than $k_B T \ln 5$ would contradict Landauer’s principle as applied to the GTE five-sector structure.

Remark 5.12 (Structure of the computability gap). The level-raising map $\pi_{\text{PSC}}^{(0)} \rightarrow \pi_{\text{PSC}}^{(\infty)}$ is a *structural correspondence*, not a categorical functor. A functor would require morphism-preservation, but the computable morphisms at Level $0 \rightarrow 1$ map to non-computable morphisms at Level $2 \rightarrow 3$; no structure-preserving map exists across the computability boundary. The correspondence is therefore at the level of MDL semantics: both levels implement $\arg \min K(x \mid \text{PSC})$, and non-computability emerges in the continuum limit.

The SRRG fixed point φ provides a *numerical bridge* across this computability gap: it is simultaneously the real-diagonal fixed point of p at Level 0 (Theorem 4.1, $\mathbf{A}_{\text{Lean}}$) and the self-consistent adjudication attractor of the SRRG at Level 2 (P27 [24], $\mathbf{A}_{\text{Lean}}$). The polynomial carries the adjudication answer in its algebraic structure, even though computing that answer in the full quantum case requires non-computable adjudication.

Remark 5.13 (Significance of Three-Level Unification). The significance of the Three-Level MDL Unification is not merely structural elegance. The same MDL principle that selects the theory from an astronomically large rule space also governs how that theory evolves as a field and how it adjudicates quantum events. These are not three separate applications of an information-theoretic principle — they are three domains in which a self-contained universe necessarily applies the same self-description criterion. The computability gap between Level $0 \rightarrow 1$ (computable PSC-restriction) and Level $2 \rightarrow 3$ (non-computable transputation) is not an inconsistency — it is the formal content of the Physical Incompleteness Theorem (analogous to Gödel’s incompleteness [6] and Turing’s undecidability [28]): a universe powerful enough to describe itself cannot compute all of its own event outcomes.

Remark 5.14 (The Level-3 residual classified: the cosmological constant as the adjudicator’s unfinalizable ledger). The same $\arg \min K$ adjudicator whose diagonal restriction has degree $\mathbf{0}'$ (Theorem 5.7) leaves a degree- $\mathbf{0}'$ residual in the Level-3 action. The irreducible residual D_{res} of the PMDL action [19] is the halting-weighted MDL sum over the realized diagonal record fragment — a left-c.e. real of Turing degree exactly $\mathbf{0}'$ (conditional on the premise bundle of Definition 5.4 together with record-measure additivity and positive witness costs; the no-computable-modulus core is unconditional, as in Theorem 5.8). The dark-energy fraction therefore inherits the transputation classification: Ω_{Λ} is limit-computable but not computable, and the two structural routes of the cosmology paper [15] — the PSC census capacity (upper) and the holographic geometric floor (lower) — are computable bracket evaluations of a value that no finite computation outputs. The cosmological constant is the adjudicator’s unfinalizable ledger at Level 3. This completes the three-level computability story: one one-jump residual, two physical shadows — measurement non-certifiability at the discrete adjudication layer (Theorem 5.8) and dark energy at the continuum field level.

6 DSAC as $[D]$ -Member and the Adjudication Hierarchy

6.1 The $[D]$ Adjudicator Class

The GTE-Möbius architecture $(A, e, [D])$ [18] designates $[D]$ as the equivalence class of all adjudication measures D satisfying the D_1 – D_5 constraints:

D_1 Internal decidability: D operates from inside the record-bearing system.

D_2 PSC-consistency: D selects only PSC-admissible trajectories.

D_3 Non-computability: D is not a total-effective algorithm on the diagonal-capable fragment.

D_4 Unique minimum: D produces a single definite outcome.

D_5 Born consistency: outcome probabilities satisfy $P(k) = |c_k|^2$.

Any two measures satisfying D_1 – D_5 are in the same PSC-coherence class $[D]$ (P48 [13]).

6.2 DSAC as a $[D]$ -Member

Differential Self-Adjudicative Computation (DSAC, [16]) is explicitly identified in P48 as “one concrete member of $[D]$ ”: DSAC’s Δ -functional minimization satisfies D_1 – D_5 by construction (NEMS Paper 77, **A/D**). The adjudication hierarchy is therefore:

$$p \xrightarrow{\pi_{\text{PSC}}^{(0)}} f_{\text{MDL}} \xrightarrow{\text{continuum PSC-min}} \text{DSAC} \in [D] \xrightarrow{\text{non-comp. limit}} P_D^\top, \quad (10)$$

where:

- $p \rightarrow f_{\text{MDL}}$: discrete, computable PSC-projection (Proposition 5.3).
- $f_{\text{MDL}} \rightarrow \text{DSAC}$: continuum extension (**B**; formal derivation open, Open Problem 1).
- $\text{DSAC} \rightarrow P_D^\top$: non-computability emerges from D_3 constraint.

Remark 6.1 (DSAC’s role in the hierarchy). DSAC is the continuum PSC-minimizing dynamics (the PR-0 field substrate [20] with Δ -functional minimization). It occupies the intermediate position between the discrete, computable f_{MDL} projection and the continuous, non-computable $P_D^\top$ adjudication. The full chain (10) provides the complete picture of MDL-minimization at increasing physical scales.

Remark 6.2 (DSAC as the physical stage sequence). The classification of Section 5.3.1 gives DSAC’s position precise computability-theoretic content. The computable stage sequence g of Theorem 5.6 is exactly what DSAC’s relaxation-to-coherence dynamics implements physically: between witness events the dissonance functional decreases monotonically toward the current sector argmin, and each arriving Σ_1 witness triggers a bounded revision (a mind change in the formal sense). The adjudication outcome is the limit of this physical relaxation — Putnam trial-and-error computation [9] in the literal sense, with the stage index realized as physical relaxation time. What no physical implementation provides is a certificate of convergence: by Theorem 5.8, “the relaxation has settled” is not an internal observable. The stage-convergence structure, the mind-change bound, and the dissonance Lyapunov behavior are verified computationally (**A**, Appendix B).

7 The Discrete Measurement Mechanism

The preceding sections assemble into a complete formal account of discrete quantum measurement in the GTE framework:

Pre-measurement: Class 3 dynamics. The quantum state evolves under p -dynamics on $\text{GF}(7)^5$. As shown in Theorem 3.4, 99.69% of initial states asymptote to the period-475 attractor; none of these trajectories pass through the PSC-admissible SM generation orbit. The state is a superposition over PSC-admissible p -trajectories.

Transputation fires: PSC-projection selects the orbit. The adjudicator $P_D^\top$ applies π_{PSC} — the MDL-minimal selection over all trajectories consistent with the macroscopic record w . By D_2 , only PSC-consistent trajectories are selected; by D_4 , a unique minimum is reached; by D_5 , the probability of each outcome k is $|c_k|^2$.

Post-measurement: the f_{MDL} orbit. The post-measurement state is on the f_{MDL} orbit $\text{GEN}_1 \rightarrow \text{GEN}_2 \rightarrow \text{GEN}_3 \rightarrow \text{VAC}$, the unique PSC-admissible trajectory (`fmdl_z7_three_generation_orbit`, $\mathbf{A}_{\text{Lean}}$). This is the realized physical sequence: the three SM generations decaying to vacuum. GEN_1 is a Garden of Eden (`fmdl_gen1_is_garden_of_eden`, $\mathbf{A}_{\text{Lean}}$), so the orbit has no predecessor — it is the unique origination point of the physical sequence.

Theorem 7.1 ($\mathbf{A}_{\text{Lean}}$ — `fmdl_orbit_is_unique_psc_trajectory`). *Under deterministic f_{MDL} dynamics (`fmdl_step5`), the trajectory $\text{GEN}_1 \rightarrow \text{GEN}_2 \rightarrow \text{GEN}_3 \rightarrow \text{VAC}$ is the unique three-step PSC-admissible orbit from GEN_1 : at each stage the successor state is uniquely determined, GEN_1 has no predecessor (Garden of Eden), and no alternative three-step PSC-consistent sequence exists. Machine-certified in *Universality/CUP3DUniqueness.lean*, zero sorry.*

Summary.

Theorem 7.2 (Discrete Measurement Mechanism, $\mathbf{A/D}$). *In the GTE framework, quantum measurement proceeds in three stages:*

- (i) *The pre-measurement quantum state is a superposition over p -compatible trajectories in $\text{GF}(7)^5$.*
- (ii) *Transputation $P_D^\top \in [D]$ applies π_{PSC} , selecting the unique PSC-admissible trajectory and producing a definite outcome with probability $|c_k|^2$.*
- (iii) *The post-measurement state lies on the $\text{GEN}_1 \rightarrow \text{GEN}_2 \rightarrow \text{GEN}_3 \rightarrow \text{VAC}$ orbit, the unique fixed point of the f_{MDL} dynamics (Theorem 7.1, $\mathbf{A}_{\text{Lean}}$).*

This mechanism is non-algorithmic (Proposition 5.3), lawful (D_1 – D_5 constraints), and fully internal to the PSC-closed system.

Remark 7.3. The Hilbert space structure, the spectral content, and the full Born rule derivation from information-loss on the f_{MDL} ring are established in P37 [23], which this theorem complements. P37 derives the measurement rule from the information-theoretic structure of the ring; the present paper identifies the *discrete CA mechanism* through which transputation selects the PSC-admissible trajectory. The two perspectives are mutually consistent and address complementary levels of description.

Remark 7.4 (Transputation as the GTE quantum computation mechanism). The Φ_{MDL} kink superselection sectors fuse additively ($w_1 \otimes w_2 = w_1 + w_2 \pmod{7}$), giving an abelian $\mathbb{Z}_7$ Dijkgraaf–Witten modular tensor category with all quantum dimensions equal to 1 (P48 [13] §The Kink Fusion Category). This is not the Fibonacci MTC: GTE is therefore *not* a universal topological quantum computer via Fibonacci-anyon braiding. However, GTE implements quantum computation by a distinct mechanism: transputation. Each adjudication event is a quantum computation — an MDL-optimal selection among the five PSC-admissible sector alternatives, governed by the D_1 – D_5 constraints. The computation is informationally protected via MDL minimality (Theorem 7.2), not topologically protected via non-abelian braiding. Fault-tolerant quantum memory is available via the $\mathbb{Z}_7$ Dijkgraaf–Witten abelian toric code at code distance $d = 7$; universal topological quantum computation would require activating the F_{21} gauge holonomy in 2+1D.

8 The Measurement Problem: GTE vs. Prior Frameworks

Quantum mechanics, as ordinarily formulated, requires an external rule — the Born rule together with wavefunction collapse — grafted onto unitary Schrödinger evolution. Unitary dynamics alone never selects a unique outcome; something outside the formalism must be invoked. This is the *measurement problem*.

Copenhagen / Bohr. The Copenhagen interpretation defines measurement by reference to a classical apparatus, without accounting for the quantum–classical boundary. In GTE, that boundary *is* the Level $0 \rightarrow 1$ transition ($p \rightarrow f_{\text{MDL}}$): the “classical apparatus” corresponds to the PSC-admissible orbit filter.

Many-worlds / Everett. Everettian quantum mechanics [4] eliminates collapse: all branches exist, with the Born rule supplied by decision theory or branch-counting arguments rather than derivation. GTE’s response: transputation is the unique PSC-forced branch selector. GTE does not branch — it selects exactly one PSC-consistent trajectory, non-computably but lawfully.

Bohmian mechanics / de Broglie–Bohm. Bohmian mechanics [1] assigns particles definite positions guided by a pilot wave; the theory is explicitly non-local. In GTE, the f_{MDL} orbit is the deterministic trajectory (deterministic CA dynamics); the pilot-wave role is played by the PSC-admissible sector of Φ_{MDL} . GTE is local (AFCA causal invariance); Bohm is not.

Objective collapse / GRW. GRW-type theories [5] postulate physical wavefunction collapse triggered by a stochastic process. In GTE, transputation is a physical collapse-like process, but forced by PSC (non-algorithmic, lawful) rather than postulated. The discrete mechanism ($p \rightarrow f_{\text{MDL}}$ orbit) is GTE’s derived collapse.

Decoherence approaches. Decoherence [30] explains apparent classicality and selects a preferred basis, but does not resolve the measurement problem: the Born rule remains an external postulate. In GTE, the Born rule is derived (four routes, $\mathbf{A}_{\text{Lean}}$). Decoherence corresponds to the Level 0 chaotic dynamics of p ; the transition to f_{MDL} dynamics is the decoherence event that resolves outcomes.

GTE’s position. Transputation is the fourth option established in P48 [13]: (a) lawful (D_1 – D_2), (b) non-algorithmic (D_3 , forced by the Physical Incompleteness Theorem), (c) PSC-consistent (D_4), and (d) Born-consistent (D_5). It differs from collapse (no external rule), many-worlds (selects, not branches), Bohm (local, not guided), and GRW (derived, not postulated).

What Transputation Is Not

Transputation (the GTE measurement mechanism) is *not*:

- **Copenhagen collapse:** no external classical apparatus is invoked; the quantum-classical boundary IS the Level $0 \rightarrow 1$ transition ($p \rightarrow f_{\text{MDL}}$).
- **Many-worlds branching:** exactly one PSC-consistent trajectory is realized, not all.
- **Bohmian pilot wave:** GTE is local (AFCA causal invariance); no non-local hidden variable.
- **GRW objective collapse:** transputation is derived from PSC, not postulated as an additional law.
- **Decoherence alone:** decoherence selects a preferred basis but does not resolve outcomes; transputation resolves outcomes.
- **Hypercomputation:** the adjudication function on the diagonal fragment has Turing degree exactly $\mathbf{0}'$ and delivers no certified Σ_1^0 decisions at finite stages; physics never requires oracle access above the first jump.
(Theorems 5.7 and 5.8)

Transputation IS: the unique PSC-consistent, non-computable, lawful internal adjudicator forced by the Physical Incompleteness Theorem (`closed_choice_forces_transputation`, $\mathbf{A}_{\text{Lean}}$, `transputation-lean`).

9 Open Problems

Open Problem 1 (Continuum limit $f_{\text{MDL}} \rightarrow \text{DSAC}$). Derive formally that DSAC is the continuum limit of the f_{MDL} PSC-projection, in the sense that the DSAC Δ -functional on Φ_{MDL} field configurations reduces to the discrete PSC-restriction $\pi_{\text{PSC}}^{(0)}$ in the lattice limit. Current status: $\mathbf{B}$ (plausible from the PR-0 field substrate structure [20], no formal derivation).

Open Problem 2 (DSAC D_3 non-computability). Prove that DSAC satisfies the D_3 constraint: DSAC cannot decide its own convergence on diagonal-capable input structures. The conditional analytic content is now available: any $[D]$ -member realizing the stage sequence of Theorem 5.6 admits no computable convergence modulus (Theorem 5.8), so DSAC inherits the D_3 property conditional on the premise bundle of Definition 5.4 and on DSAC’s identification with that stage sequence (Remark 6.2). What remains open is the machine certification of DSAC’s membership itself, which would upgrade it from $\mathbf{A}/\mathbf{D}$ (citation-based) to $\mathbf{A}_{\text{Lean}}$. Lean candidate: `dsac_member_of_D_class`.

10 Conclusion

The central result of this paper is the Three-Level MDL Unification: a single MDL principle governs theory selection (p from rule space), field dynamics (Φ_{MDL} via PMDL), and quantum event adjudication (Born rule via transputation). The discrete measurement mechanism is established at $\mathbf{A}/\mathbf{D}$, with orbit uniqueness now machine-certified at $\mathbf{A}_{\text{Lean}}$ (Theorem 7.1). The companion result is the two-sided computability classification: under the named premise bundle PR1–PR5, the adjudication function on the diagonal fragment has Turing degree exactly $\mathbf{0}'$ — quantum measurement is Gold–Putnam trial-and-error computation at the halting degree, never above the first jump, with no computable convergence modulus (Section 5.3.1). The same one-jump residual classifies the cosmological constant as a left-c.e. real of degree $\mathbf{0}'$ (Remark 5.14). The remaining gaps are well-posed mathematical questions — the continuum limit $f_{\text{MDL}} \rightarrow \text{DSAC}$ (Open Problem 1) and DSAC D_3 certification (Open Problem 2) — not gaps in the framework. The cross-module Lean bundle for Three-Level Unification is now machine-certified (§5.2, `mdl_three_level_unification`).

This paper completes the polynomial certificate chain begun in P49 [21], situating the measurement mechanism relative to P37 [23] (Hilbert space and Born rule) and P48 [13] (transputation framework and MDL three roles). The GTE polynomial p is the connecting object: its PSC-projection f_{MDL} defines discrete measurement; its real diagonal encodes the SRRG fixed point; and it enters the PMDL action at continuum scale.

A Lean Certification Table

All Lean certifications cited in this paper. Theorems live in three public repositories: `ugp-lean` (canonical), `transputation-lean`, and `nems-lean`; all entries are zero sorry. Entries marked † are conditional certificates: their physics premises enter as named Lean hypothesis structures (explicit structure fields, never bare axioms), as disclosed after the table.

Table 1: Lean 4 certifications cited in this paper. All entries are zero sorry. UL = ugp-lean; TL = transputation-lean; NL = nems-lean.

Theorem name	Module	Repo	Cat
psc_projection_gives_fmdl	Polynomial/ PolyExplorations	UL	A _{Lean}
fmdl_psc_projection_of_p	Polynomial/ PolyExplorations	UL	A _{Lean}
fmdl_zero_on_free_ neighborhoods	Universality/ CUP3DUniqueness	UL	A _{Lean}
p_poly_agrees_fmdl_on_binary	Universality/ Z7Invariant Subsets	UL	A _{Lean}
p_fmdl_disagree_on_orbit	Polynomial/ PolyExplorations	UL	A _{Lean}
kl_divergence_fmdl_p_nonzero	Polynomial/ PolyExplorations	UL	A _{Lean}
fmdl_z7_three_generation_ orbit	Universality/ CUP3DUniqueness	UL	A _{Lean}
fmdl_gen1_is_garden_of_eden	Universality/ CUP3DUniqueness	UL	A _{Lean}
fmdl_orbit_is_unique_psc_ trajectory	Universality/ CUP3DUniqueness	UL	A _{Lean}
mdl_three_level_orbit_bundle	Universality/ CUP3DUniqueness	UL	A _{Lean}
mdl_three_level_polynomial_ bundle	Universality/ CUP3DUniqueness	UL	A _{Lean}
mdl_three_level_unification	Polynomial/ MDLThreeLevel Unification	UL	A _{Lean}
mdl_level23_closed_choice_ forces_transputation	Polynomial/ MDLThreeLevel Unification	UL	A _{Lean}
period_475_returns	Polynomial/ PolyExplorations	UL	A _{Lean}
period_475_is_minimal	Polynomial/ PolyExplorations	UL	A _{Lean}
gte_poly_srrg_bridge	Algebra/ SRRGCABridge	UL	A _{Lean}
srrg_equals_mdl_minimization	Algebra/ SRRGCABridge	UL	A _{Lean}
unique_cubic_gravity_coupling	Gravity/ PMDLGravity Theorems	UL	A _{Lean}
closed_choice_forces_ transputation	Theorems/Forced Adjudication	TL	A _{Lean}
halting_manyOne_reducible_to_ RT	Diagonal/Sigma1 Completeness	NL	A _{Lean}

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Theorem name	Module	Repo	Cat
rt_manyOne_reducible_to_halting_zero	Diagonal/Sigma1 Completeness	NL	$\mathbf{A}_{\text{Lean}}$
rt_sigma1_complete_on_diagonal	Diagonal/Sigma1 Completeness	NL	$\mathbf{A}_{\text{Lean}}$
no_computable_convergence_modulus	Diagonal/ NoConvergence Modulus	NL	$\mathbf{A}_{\text{Lean}}$
pt_diagonal_manyOne_degree_halting	Theorems/ DiagonalDegree	TL	$\mathbf{A}_{\text{Lean}}$
pt_limit_computable	Theorems/Limit Computability	TL	$\mathbf{A}_{\text{Lean}}^{\dagger}$
pt_diagonal_degree_halting	Theorems/ DiagonalDegree	TL	$\mathbf{A}_{\text{Lean}}^{\dagger}$
born_rule_unconditional	Universality/ PSEffectMeasure	UL	$\mathbf{A}_{\text{Lean}}$
landauer_heat_floor_transputation	Measurement/ LandauerFloor	UL	$\mathbf{A}_{\text{Lean}}$
landauer_heat_ceiling_transputation	Measurement/ LandauerFloor	UL	$\mathbf{A}_{\text{Lean}}$
mdl_tower_bundle	Framework/ MDLTower	UL	$\mathbf{A}_{\text{Lean}}$
mdl_tower_three_levels_non_circular	Framework/ MDLTower	UL	$\mathbf{A}_{\text{Lean}}$
mdl_ca_rule_coding_closed	Universality/ MDLDerivability Criterion	UL	$\mathbf{A}_{\text{Lean}}$
ugp_orbit_interpolation_lift	Universality/ TriangleLift Theorem	UL	$\mathbf{A}_{\text{Lean}}$
K_CA_CMCA	Framework/CMCAML Minimality	UL	$\mathbf{A}_{\text{Lean}}$
asr_rt_not_computable	Diagonal/ HaltingReduction	NL	$\mathbf{A}_{\text{Lean}}$

Conditionality disclosure for the computability-classification certificates. The premise bundle of Definition 5.4 enters the Lean formalization as named hypothesis structures, never as bare axioms: PR1 is the structure `Sigma1RecordTruth`, PR5 is `RecordReadout` (both `nems-lean`, `Diagonal/Premises`), and PR2–PR4 are the fields of `AdjudicationPremiseBundle` (`transputation-lean`, `Core/Premises`). The unmarked $\mathbf{A}_{\text{Lean}}$ entries are unconditional given those premise structures as stated in the corresponding theorems (`halting_manyOne_reducible_to_RT` requires no premise at all). The two entries marked † carry an additional layer of conditionality that must be disclosed precisely: `pt_limit_computable` takes the stage-convergence property itself as the named structure `StageConvergence` (the structure supplies the computable stage function and its convergence; the Lean theorem packages the consequence rather than deriving convergence from PR1–PR4, which is the analytic content of Theorem 5.6, **A/D**); and `pt_diagonal_degree_halting` bundles `StageConvergence` together with the mind-change bound as the named structure `MindChangeBounded`. The full Turing-degree formalization of Theorem 5.7 (replacing the certified many-one

form `pt_diagonal_manyOne_degree_halting` by Turing reducibility) remains open pending Turing-reduction infrastructure in Mathlib.

B Code and Reproducibility

The KL-divergence computation, structural analysis, and stage-convergence experiments from which the numerical results in Sections 3.2 and 5.3.1 are drawn are implemented in the scripts under `papers/51_polynomial_transputation/scripts/`:

- `p_transputation_kl_divergence.py` — Computes KL divergence, entropy, and information-theoretic gap between p and f_{MDL} as distributions over $\text{GF}(7)^3$; verifies the diagonal fixed-point φ .
- `p_transputation_formal_analysis.py` — Functor analysis, improved f_{MDL} model with orbit windows, and the Three-Level MDL correspondence.
- `transputation_limit_stage_convergence.py` — Stage-convergence structure of $[D]$ -adjudication (Section 5.3.1): the diagonal halting-coded family (one mind change; convergence stage equal to halting time), the compound-record mind-change bound (500 trials, $k = 5$ winding sectors, maximum 3 vs. bound 8), the DSAC-style Lyapunov dissonance relaxation (Remark 6.2), and the no-computable-modulus phenomenology (Theorem 5.8). Produces `transputation_limit_stage_convergence_results.json`.

All scripts are self-contained, include wall-clock timeouts, and produce JSON artifacts with all numerical outputs (`p_transputation_kl_results.json` for the first two scripts). The scripts contain no local paths; all inputs are relative to the script directory.

To reproduce the distributional KL-divergence results from Section 3.2:

```
cd papers/51_polynomial_transputation/scripts
python3 p_transputation_kl_divergence.py
```

Expected output: $\text{KL}(f_{\text{MDL}} \parallel p)$ in the range $[4.6, 7.7]$ bits (Model A: 7.71 bits; Model B: 4.62 bits).

Lean verification. All theorems in Appendix A can be verified by building the four public repositories:

```
cd /path/to/ugp-lean && lake build UgpLean
cd /path/to/srrg-lean && lake build
cd /path/to/nems-lean && lake build
cd /path/to/transputation-lean && lake build
```

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